

1 Explainability in Reinforcement Learning

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Chapter 1

Experiments

This chapter presents the experimental study of the Clustering Agent Representation (CAR) method introduced in the previous chapter. It focuses on evaluating the method’s ability to consistently identify meaningful conceptual structures. The remainder of this chapter is organized as follows:

- Section 1.1 describes the software implementation of the CAR pipeline.
- Section 1.2 details the experimental configuration, including the environments, the reinforcement learning setup, and the training of the surrogate policies.
- Section 1.3 presents the training of the initial and surrogate policies and evaluates their behavioral fidelity.
- Section 1.4 analyzes the latent representations obtained by applying CAR and its metrics.
- Section ?? summarizes the findings and discusses their implications.

1.1 Software Implementation

For Deep Learning (DL) and Reinforcement Learning (RL), theoretical results are often difficult to apply to real-world cases. The effectiveness of methods in these fields is therefore primarily assessed through experimentation. Consequently, particular attention is given to the implementation of CAR. It relies on established libraries that provide efficient implementations of its main components. The code is made publicly available as open-source to support reproducible replication of the reported results.¹

CAR is organized as a pipeline of five stages, shown in Figure 1.1. Each stage is an independent executable, and stages communicate through artifacts written to disk. This allows each stage to be rerun or inspected independently. The parameters of each experiment are gathered in a single configuration object, described in Section 1.2. The following subsections describe the purpose of each stage, its implementation, and the libraries used.

¹https://github.com/Maxalaar/Clustering_Agent_Latent_Space

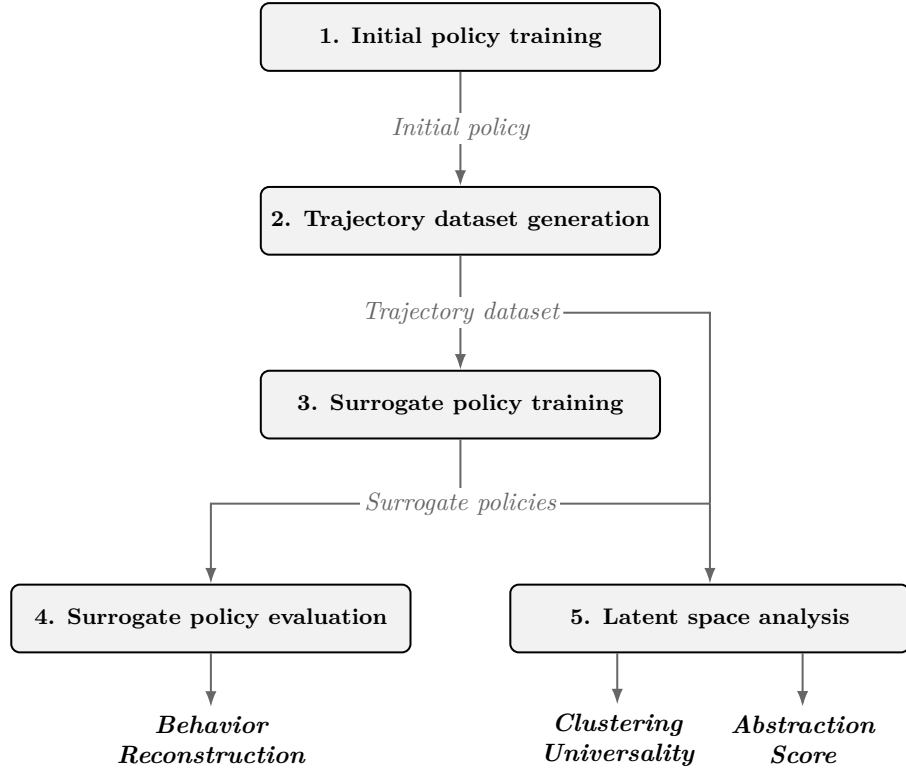


Figure 1.1: The CAR pipeline.

1.1.1 Initial policy training

The first stage trains the initial policy using RLlib [Moritz et al., 2018, Liang et al., 2018]. RLlib provides scalable implementations of RL algorithms and supports distributed training. The use of RLlib also allows CAR to be applied to environments implemented using the Gymnasium interface [Towers et al., 2024]. By default, the initial policy is trained using Proximal Policy Optimization (PPO) [Schulman et al., 2017]. Other RLlib algorithms, such as Asynchronous Proximal Policy Optimization (APPO) [Espeholt et al., 2018], Deep Q-Networks (DQN) [Mnih et al., 2013], and DreamerV3, can also be selected through the same configuration.

The policy and critic networks are implemented using PyTorch [Paszke et al., 2019]. PyTorch is the standard deep learning frameworks and is widely used for developing and training neural networks. It provides GPU acceleration and supports a wide range of neural network architectures. This makes it well suited for implementing the different policy and critic architectures used in the experiments.

The resulting policy and critic parameters are saved as a checkpoint together with the environment and training configurations.

1.1.2 Trajectory dataset generation

The second stage builds the dataset used to train the surrogate policies. RLlib is used to collect trajectories in parallel using multiple environment runners. For each step, the dataset stores the observation together with the parameters of

the action distribution produced by the initial policy.

Trajectories are stored as Parquet files [Apache Arrow, 2016]. Its columnar structure allows efficient data access and supports concurrent processing. This is useful here because the datasets contain around one million observation–action pairs and cannot be loaded entirely into memory.

The resulting dataset is used both to train the surrogate policies and to generate the latent representations analyzed by CAR.

1.1.3 Surrogate policy training

The third stage trains the surrogate policies using behavior cloning on the trajectory dataset. This provides a supervised learning approach for replicating the initial policy. The training procedure is implemented using PyTorch Lightning [Falcon, 2019]. PyTorch Lightning provides a simplified interface for training neural networks with PyTorch. It handles common training operations such as data loading, optimization, checkpointing, and GPU execution.

Each surrogate policy is saved as a checkpoint together with the configuration used for training.

1.1.4 Surrogate policy evaluation

The fourth stage evaluates how faithfully the surrogate policies reproduce the behavior of the initial policy. Each surrogate is wrapped as an RLlib policy and evaluated in the original environment. Its mean cumulative reward over multiple episodes is compared with that of the initial policy. Evaluation episodes are collected using parallel environment runners.

This stage computes the *Behavior Reconstruction*. It therefore verifies that the surrogate policies reproduce the behavior of the initial policy sufficiently well before their latent representations are analyzed. If a surrogate does not meet the required level of behavior reconstruction, its training can be repeated.

1.1.5 Latent space analysis

The fifth stage performs the main analysis of CAR. The trajectory dataset and trained surrogate policies are used to generate the latent representations. These representations form the point clouds that are analyzed using clustering algorithms.

The clustering step relies on the cuML library [RAPIDS, 2023], which provides GPU-accelerated clustering algorithms. This reduces the computational cost and allows point clouds containing tens of thousands of representations to be analyzed.

The resulting partitions are used to compute the *Clustering Universality* and the *Abstraction Score*. The stage also produces the material required for the qualitative analysis, including the clustered point clouds and the cluster assignment of each observation.

Table 1.1 summarizes the main third-party libraries used in the pipeline.

Pipeline Stage	Library	Description
Initial policy training	RLlib [Moritz et al., 2018, Liang et al., 2018]	Distributed RL library used to train the initial policy.
	PyTorch [Paszke et al., 2019]	Deep learning framework implementing the policy and critic networks.
	Gymnasium [Towers et al., 2024]	Common interface used to plug in any RL environment.
Trajectory dataset generation	RLlib [Moritz et al., 2018, Liang et al., 2018]	Distributed environment runners used to collect trajectories in parallel.
	Apache Arrow [Apache Arrow, 2016]	Columnar file format (Parquet) used to store the trajectory datasets.
Surrogate policy training	PyTorch Lightning [Falcon, 2019]	Structured training loop used for the behavior cloning of the surrogate policies.
Surrogate policy evaluation	RLlib [Moritz et al., 2018, Liang et al., 2018]	Wraps each surrogate policy as an RL policy to evaluate it in the environment.
Latent space analysis	cuML [RAPIDS, 2023]	GPU-accelerated clustering algorithms used to analyze the latent representations.

Table 1.1: Main software libraries used at each stage of the CAR pipeline.

1.2 Experimental Configuration

This section describes the configuration of the experiments reported in this chapter. It covers the environments, the neural architecture of the policies, and the main parameters of each pipeline stage.

1.2.1 Environments

All experiments presented in this manuscript use environments from the Atari benchmark [Bellemare et al., 2013], accessed through the Arcade Learning Environment. The use of Atari environments offers several advantages. First, Atari is a widely used benchmark in the RL community, with many studies using these environments as standard evaluation settings. This makes the experimental results easier to compare with existing work. Second, these environments were not specifically designed for RL, which reduces the risk of introducing biases related to the design of the environments themselves. Finally, the benchmark provides a diverse set of well-known environments with varying levels of complexity. This diversity makes it possible to evaluate CAR across different types of tasks and levels of difficulty.

The configuration of the Atari environments follows common practice in RL research [Machado et al., 2017].

In each environment, the agent receives visual observations in the form of raw pixels. Each image is resized to 64×64 pixels and converted to grayscale to reduce memory requirements. To provide information about the dynamics of the environment, four consecutive observations are stacked along the channel dimension. The resulting observation has a shape of $64 \times 64 \times 4$. This allows the agent to infer the motion of objects from successive observations rather than relying on a single frame.

Frame skipping is also used to reduce the duration of each episode. The agent selects a new action every five frames, while the selected action is repeated for

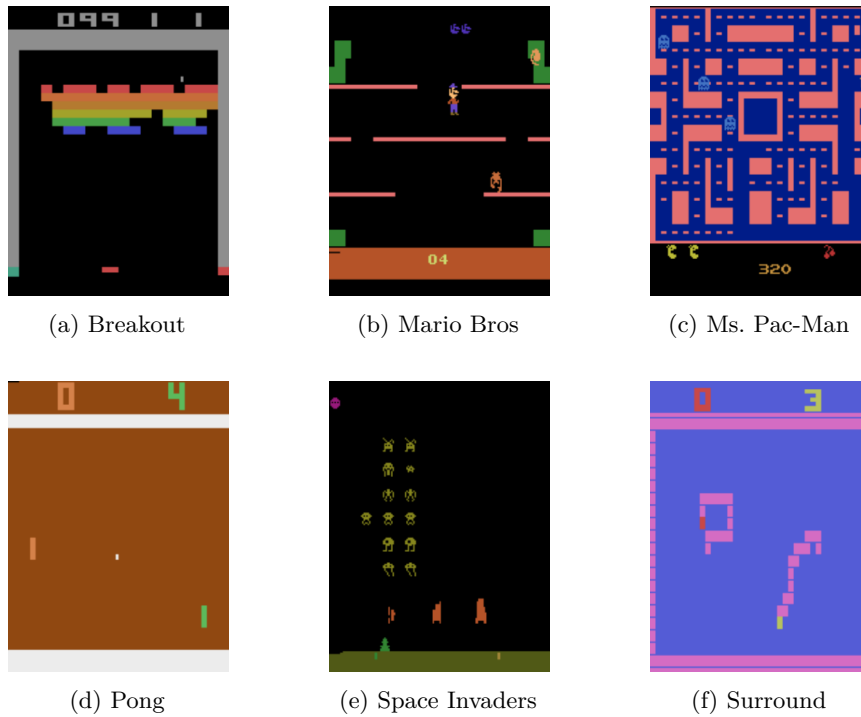


Figure 1.2: The six Atari environments used to evaluate CAR.

the intervening frames. This provides sufficient temporal resolution for effective control while reducing the number of decisions required during an episode.

The actions correspond to the different buttons available on the Atari controller. The resulting action space is therefore discrete. Since Atari games are fully deterministic, sticky actions are introduced to add stochasticity to the trajectories. With a probability of 0.05, the environment repeats the action selected at the previous step instead of applying the new action. This prevents the agent from simply memorizing a fixed sequence of actions during training.

With the environment configuration established, the objective of each environment is described below. Figure ?? provides an overview of the six environments used in the experiments.

Breakout. The agent moves a paddle at the bottom of the screen (Figure 1.2a) to bounce a ball into a wall of bricks, each broken brick scoring a point, and the ball speeding up as the wall is cleared.

Mario Bros. A single-screen platform game (Figure 1.2b). The player clears each phase by hitting enemies from below to flip them, then kicking them off the platforms before they recover. A central POW block flips every enemy at once but can only be used a few times per phase.

Ms. Pac-Man. The player moves through a maze (Figure 1.2c), eating pellets while avoiding four ghosts, with power pellets briefly making the ghosts edible. The maze structure and the many possible local situations, such as a nearby

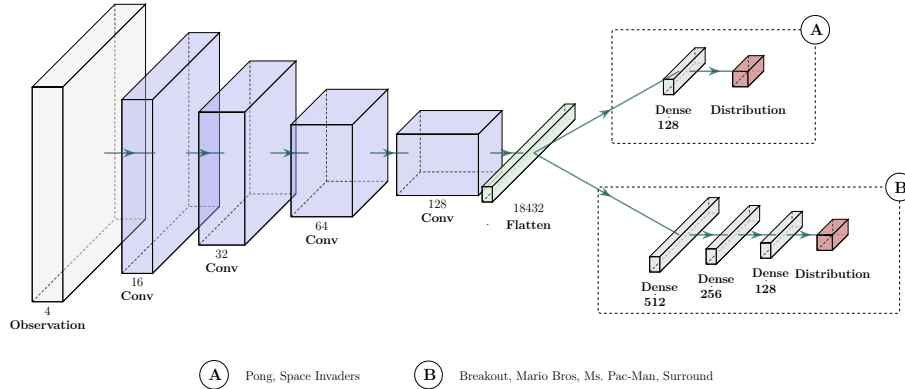


Figure 1.3: Architecture of the surrogate policies.

ghost, an intersection, or an active power pellet, produce a large number of distinguishable states.

Pong. A minimal two-player game (Figure 1.2d) in which the agent controls one paddle and must send the ball past the opponent.

Space Invaders. The agent moves a cannon along the bottom of the screen (Figure 1.2e), firing at descending rows of aliens while dodging their shots and using destructible shields, with a bonus saucer occasionally crossing the top.

Surround. A two-player game on a grid (Figure 1.2f). Each player leaves a solid trail behind a moving cursor, and the goal is to force the opponent to crash into a trail or a wall.

1.2.2 Neural architecture

The surrogate policies use a convolutional Neural Network (NN) to process the visual observations. The convolutional part consists of four layers with 16, 32, 64, and 128 channels, using 4×4 kernels with a stride of 2. Each convolutional layer is followed by a LeakyReLU activation. The resulting representation is then flattened and passed through a multilayer perceptron that produces the parameters of the action distribution.

The architecture of the multilayer perceptron varies across environments. For Pong and Space Invaders, it contains a single hidden layer with 128 units. For the other four environments, it contains three hidden layers with 512, 256, and 128 units. The size of the output layer matches the number of actions available in each environment: 4 for Breakout, 18 for Mario Bros, 9 for Ms. Pac-Man, 6 for Pong, 6 for Space Invaders, and 5 for Surround.

Figure 1.3 illustrates both configurations.

1.2.3 Parameters

Having defined the environments and neural architectures, the remaining parameters of the CAR pipeline must now be specified. The main parameters

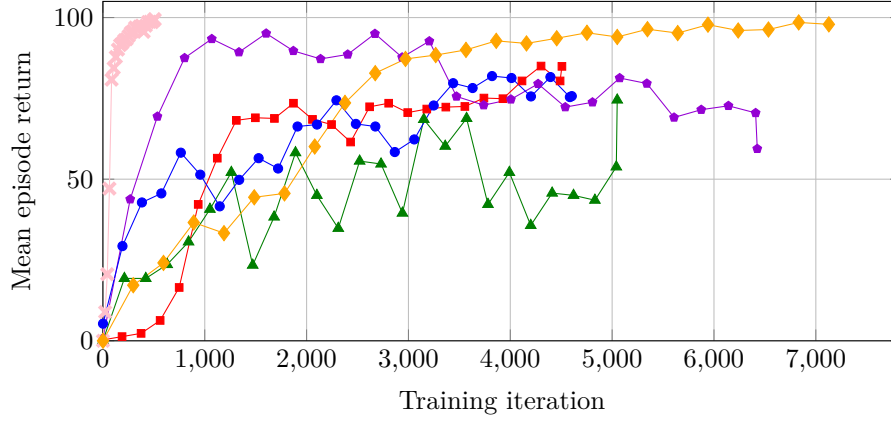
Environment configuration	
Frame size	64×64
Frames stacked per observation	4
Frame skip	5
Sticky-action probability	0.05
Initial policy training	
Algorithm	PPO
Optimizer	Adam
Learning rate	1.5×10^{-4}
GAE λ	0.95
Policy clipping ε	0.1
Value-function clipping	10
Entropy coefficient	0.01
KL coefficient	0.5
Epochs per update	10
Gradient clipping (global norm)	100
Surrogate policy training	
Surrogate policies per environment	3
Observation–action pairs per dataset	$\geq 10^6$
Batch size	2048
Optimizer	Adam
Learning rate	1×10^{-4}
Latent space analysis	
Embedding per surrogate policy	50 000
Clustering algorithm	k -means
Distance	Euclidean distance
Number of clusters K	2 to 30

Table 1.2: Main experimental parameters.

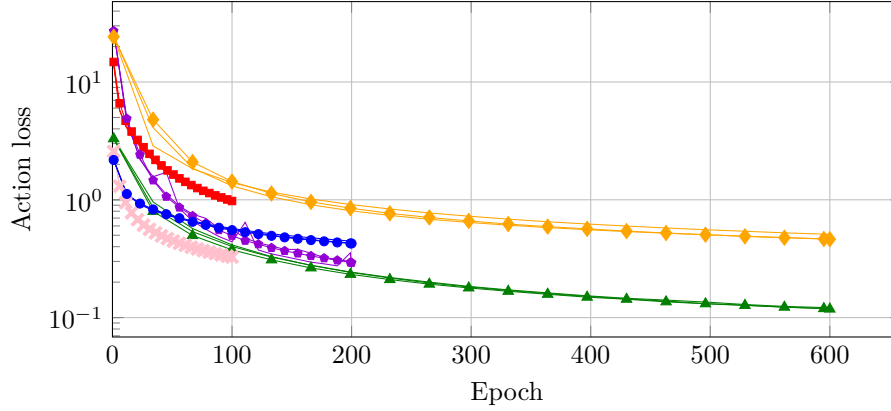
shared across the experiments are summarized in Table 1.2. There are too many parameters to discuss them all in detail.

Given the large number of possible configurations, it is not feasible to evaluate all possible combinations. The choices made in this work are therefore based primarily on standard practices from the literature and on practical experience. Whenever possible, we used the simplest suitable configuration. For the clustering algorithm and distance metric in particular, the simplest suitable choices are used to establish a baseline.

These choices do not restrict the CAR methodology itself. Most components can be replaced or configured differently depending on the application. The evaluation metrics provide an objective basis for comparing these alternatives and determining which configurations produce the most consistent and meaningful results.



(a) Training of the initial policies with PPO.



(b) Training action loss of the surrogate policies.

—▲— Breakout —■— Mario Bros —●— Ms. Pac-Man
—×— Pong —●— Space Invaders —◆— Surround

Figure 1.4: Training of the initial and surrogate policies.

1.3 Policy Training Analysis

This section reports the results of training the initial and surrogate policies for CAR, applied to the six environments. Section 1.3.1 verifies that the initial policies reach a competent level of play, while Section 1.3.2 evaluates whether the surrogate policies faithfully reproduce their behavior using the *Behavior Reconstruction*.

1.3.1 Initial policy training

Figure 1.4a shows how the initial policies improve during PPO training. For each environment, the mean episode return is normalized so that 0% and 100% correspond to the lowest and highest mean return observed during that run. In every environment the return increases and then reaches a plateau, at which

Environment	<i>Behavior Reconstruction</i>
Breakout	0.83 $\pm$ 0.03
Mario Bros	0.99 $\pm$ 0.03
Ms. Pac-Man	1.00 $\pm$ 0.03
Pong	0.95 $\pm$ 0.01
Space Invaders	0.97 $\pm$ 0.07
Surround	1.00 $\pm$ 0.01

Table 1.3: Evaluation of *Behavior Reconstruction* across the six Atari environments.

point training is stopped. The number of iterations required to reach this plateau varies from a few hundred, for Pong, to several thousand, for Surround and Ms. Pac-Man. For Ms. Pac-Man the return decreases after its peak. Since the best checkpoint is retained rather than the last one, this does not affect the policy that is subsequently analyzed.

To confirm that the trained policies play competently, their scores are compared with the reference scores reported by [Machado et al., 2017]. Each initial policy reaches a score of the same order as these references, so it constitutes a reasonable target for imitation.

1.3.2 Surrogate policy training

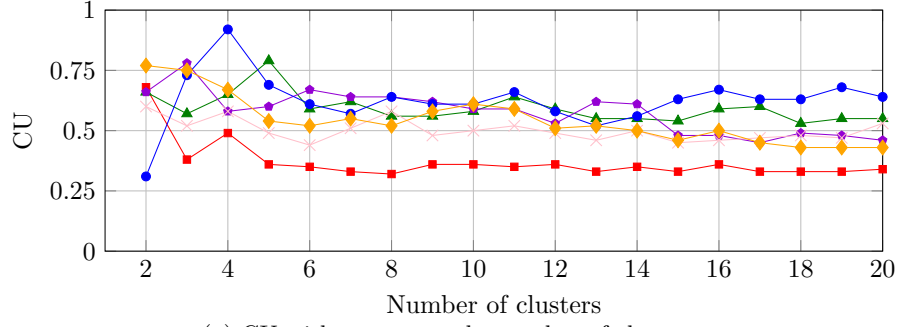
Figure 1.4b shows, for each environment, the training action loss of the three surrogate policies. In every case the loss decreases smoothly, and the three independently trained policies follow almost identical trajectories, converging to the same value.

We further evaluate the ability of the surrogate policies to replicate the behavior of the initial policies using the *Behavior Reconstruction* (Table 1.3). The surrogate policies reach a *Behavior Reconstruction* of 0.96 on average over 100 evaluation episodes, ranging from 0.83 for Breakout to 1.00 for Ms. Pac-Man and Surround. They therefore reproduce the decisions of the initial policies closely enough for their latent representations to be analyzed in place of those of the initial policies.

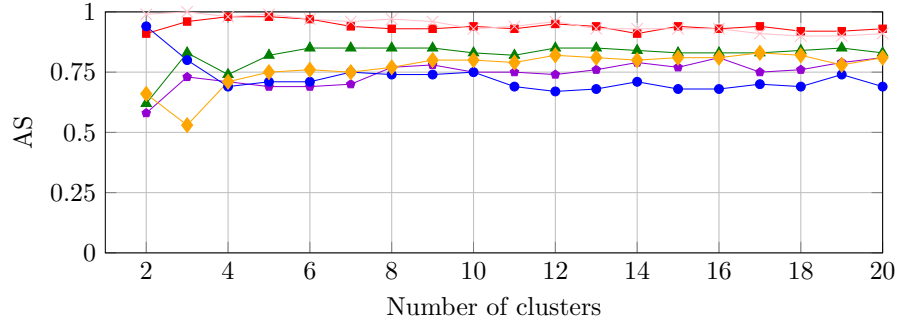
1.4 Latent Space Analysis

With the behavioral fidelity of the surrogate policies established in Section 1.3.2, this section analyzes their latent representations using Clustering Universality (CU), Abstraction Score (AS), and Adjusted Clustering Universality (ACU). Unlike the single-layer analysis reported previously, the representation extracted after every convolutional layer of the surrogate policies is now considered, so that the effect of depth on the recovered conceptual structure can be assessed. For each environment and each analyzed layer, a random subset of 50,000 observations is embedded into the Latent Space (LS) of each surrogate policy, and the resulting point cloud $\mathcal{E}$ is partitioned with k -means clustering [MacQueen, 1967] for every number of clusters $K \in [2, 20]$.

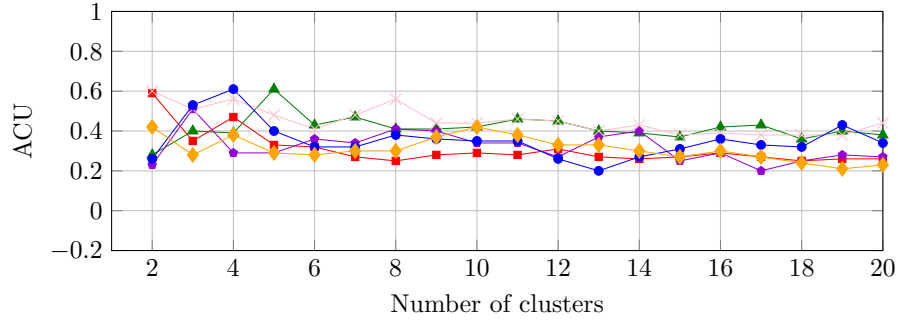
The following subsections report, for each environment, the evolution of CU,



(a) CU with respect to the number of clusters.



(b) AS with respect to the number of clusters.



(c) ACU with respect to the number of clusters.

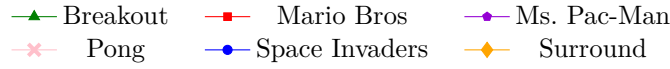


Figure 1.5: CU, AS, and ACU with respect to the number of clusters, for each of the six environments, for the layer that maximizes the average ACU in that environment.

AS, and ACU with respect to K , for the layer that maximizes the average ACU across K – i.e., the layer at which the agreement between independently trained surrogate policies is least explained by the observation space, the action space, or the network’s initialization. The full breakdown across every analyzed layer is reported in Appendix ??.

231 1.4.1 Breakout

232 For Breakout, ACU is maximized at layer 10, with $CU = 0.59 \pm 0.06$, $AS =$
 233 0.82 ± 0.05 , and $ACU = 0.41 \pm 0.06$ on average over K . As shown in Appendix ??,
 234 CU is highest in the earliest layers and decreases with depth, while AS follows
 235 the opposite trend, so that ACU peaks once the representation has become
 236 sufficiently abstract without yet losing cross-seed stability.

237 1.4.2 Mario Bros

238 For Mario Bros, ACU is maximized at the last analyzed layer (layer 14), with
 239 $CU = 0.37 \pm 0.08$, $AS = 0.94 \pm 0.02$, and $ACU = 0.31 \pm 0.08$ on average over
 240 K . As shown in Appendix ??, AS increases monotonically with depth, reaching
 241 near-saturation in the last layer, while CU decreases; ACU nonetheless keeps
 242 increasing with depth, indicating that the residual cross-seed agreement in the
 243 deepest layer is largely non-trivial.

244 1.4.3 Ms. Pac-Man

245 For Ms. Pac-Man, ACU is maximized at the last analyzed layer (layer 14), with
 246 $CU = 0.58 \pm 0.08$, $AS = 0.74 \pm 0.05$, and $ACU = 0.32 \pm 0.07$ on average over
 247 K . As shown in Appendix ??, the pattern mirrors Mario Bros: AS increases
 248 steadily with depth while CU decreases, and ACU is highest in the deepest layer.

249 1.4.4 Pong

250 For Pong, ACU is maximized at layer 7, with $CU = 0.50 \pm 0.04$, $AS = 0.95 \pm 0.03$,
 251 and $ACU = 0.45 \pm 0.07$ on average over K – the highest ACU among the six
 252 environments. As shown in Appendix ??, AS already saturates from layer 5
 253 onward, and ACU decreases again in the last analyzed layer (layer 10): unlike
 254 Mario Bros and Ms. Pac-Man, the most informative layer for Pong is not the
 255 deepest one.

256 1.4.5 Space Invaders

257 For Space Invaders, ACU is maximized at layer 7, with $CU = 0.63 \pm 0.11$,
 258 $AS = 0.73 \pm 0.06$, and $ACU = 0.35 \pm 0.09$ on average over K . As shown in
 259 Appendix ??, ACU decreases again toward the last analyzed layer (layer 10),
 260 following the same non-monotonic pattern observed for Pong.

261 1.4.6 Surround

262 For Surround, ACU is maximized at the last analyzed layer (layer 14), with
 263 $CU = 0.54 \pm 0.10$, $AS = 0.77 \pm 0.07$, and $ACU = 0.31 \pm 0.06$ on average over K .
 264 As shown in Appendix ??, AS increases with depth while CU fluctuates without
 265 a clear trend, and ACU is highest in the deepest layer.

266 1.4.7 Discussion

267 Across all six environments, CU tends to be highest in the earliest layers and
 268 to decrease with depth, while AS follows the opposite trend, increasing with

depth as the representation becomes less tied to the raw observation, the action distribution, and the network’s initialization. ACU, which combines both effects, therefore peaks once the representation is abstract enough for AS to be high, but before CU has decayed too much. For Mario Bros, Ms. Pac-Man, and Surround, this optimum is reached at the last analyzed layer, suggesting that further gains might be observed by analyzing even deeper representations. For Breakout, Pong, and Space Invaders, in contrast, ACU peaks at an intermediate layer and decreases again toward the last one, indicating that the deepest representation is not always the most informative for CAR. This confirms that the layer at which the analysis is performed is not a neutral methodological choice and should itself be selected according to ACU rather than fixed a priori to the last convolutional layer, as was done in the single-layer analysis reported in earlier work.

Acronyms

282	ACU Adjusted Clustering Universality 9–12
283	APPO Asynchronous Proximal Policy Optimization 2
284	AS Abstraction Score 9–12
285	CAR Clustering Agent Representation 1–8, 12
286	CU Clustering Universality 9–12
287	DL Deep Learning 1
288	DQN Deep Q-Networks 2
289	GPU Graphics Processing Unit 4
290	LS Latent Space 9
291	NN Neural Network 6
292	PPO Proximal Policy Optimization 2, 7, 8
293	RL Reinforcement Learning 1, 2, 4

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